

Import Duty Pass-Through to Domestic Gold Prices: Quasi-Experimental Evidence from India's Gold Market, 2024–2026

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Abstract

This paper estimates how much of India's May 2026 gold import duty hike, from 6 to 15%, passed through to domestic gold prices. Exploiting the sharp and unanticipated policy discontinuity as a quasi-experiment, this paper applies an interrupted time series (ITS) design with Newey-West HAC standard errors to a primary sample of 375 daily observations from the India Bullion and Jewellers Association benchmark price series (July 2024 – June 2026), with 346 pre-hike and 29 post-hike trading days. The full dataset comprises 929 daily price observations (824 IBJA PDF circulars and 105 BullionWorld.in gap-fills) spanning January 2022 through June 2026, within a 1,170-row weekday date spine.

The duty hike raised the domestic gold premium by Rs. 9,841 per 10 grams, representing 81.7% of the theoretical ceiling, with a 95% confidence interval of 73.3 to 90.1%. Full pass-through is formally rejected ($t = -4.26$, $p < 0.001$). Local projections confirm the adjustment was immediate on the first trading day and held on a stable plateau through Day 14 (78.2%), with no evidence of anticipatory pricing. An ARDL bounds test confirms cointegration between domestic and international gold prices in the pre-hike period; the stable local-projection plateau through Day 14 is consistent with the shift persisting rather than mean-reverting within the observation window. Placebo tests and counterfactual methods confirm the result is specific to May 13, 2026.

This paper also documents asymmetric price transmission: the duty hike passed through at 81.7% by the first trading day, while the July 2024 cut reached only 65.6% at an equivalent horizon (clean sample, July 2022–May 2026, $N = 746$). The hike therefore transmitted more completely than the cut, consistent with downward rigidity in duty pass-through. The incomplete pass-through is consistent with supply-side dampening from front-loaded inventory imported at the prior duty rate, seasonal demand weakness, and historical smuggling elasticity.

*Data and replication code are available at <https://github.com/OlixIgnacious/gold-policy-project>. The dataset contains 929 daily price observations spanning January 2022–June 2026 (824 from IBJA PDF circulars and 105 from BullionWorld.in, a licensed IBJA data republisher), within a 1,170-row weekday date spine (January 3, 2022 through June 30, 2026) that also includes 241 rows with no price observation (market holiday, closure, and archive gap dates).

Keywords: import duty pass-through, gold market, India, interrupted time series, quasi-experiment, commodity taxation

JEL Codes: E31, F13, H22, Q02

1 Introduction

India is the world's second-largest consumer of gold, importing between 700 and 900 tonnes annually and accounting for roughly a quarter of global physical demand^[1]. Gold imports are simultaneously a major source of current account pressure: in fiscal year 2026, gold import spending reached US\$71.98 billion, approximately 9% of total merchandise imports, contributing to a current account deficit of US\$30.1 billion between April and December 2025. In this context, the import duty on gold is not merely a revenue instrument. It is an active lever of macroeconomic management, deployed periodically to compress import demand and stabilise the rupee.

On July 23, 2024, the Union Budget reduced the basic customs duty on gold from 10% to 5% and the Agriculture Infrastructure and Development Cess (AIDC) from 5% to 1%, lowering the combined duty from 15% to 6%. This was the lowest duty rate on gold in over a decade. The immediate response was a sharp compression of domestic gold premiums toward import parity and a surge in demand, with the World Gold Council estimating the cut would add at least 50 tonnes to second-half 2024 demand^[2]. FY26 gold imports rose 24% year-on-year. By early 2026, with the rupee under sustained depreciation pressure and a persistent current account deficit, the government acted again. On the evening of May 12, 2026, a gazette notification restored the combined duty to 15%, effective from midnight, reversing the 2024 cut in its entirety and taking full effect at the opening of markets on May 13, 2026.

This paper asks a precise question: how much of the 9 percentage point duty increase passed

through to the domestic gold price? The answer is not obvious. In a textbook small open economy with competitive markets and free trade, a duty increase of $\tau\%$ should raise the domestic price by exactly $\tau\%$ of the import parity price, yielding 100% pass-through. But India's gold market has structural features that work against complete pass-through. Smuggling, which historically rises sharply when domestic premiums approach the ceiling implied by full pass-through, provides a competing supply channel that caps price increases^[3]. Demand destruction compresses the premium from the demand side. And short-run inventory dynamics, particularly the large volume of gold imported at the prior 6% duty rate in April 2026 (US\$5.6 billion), add a transitory supply buffer that suppresses the immediate price response. Measuring the actual pass-through, and understanding why it falls short of 100%, is the central empirical challenge this paper addresses.

This paper exploits the May 13, 2026 duty hike as a quasi-experiment. The hike was announced at midnight without legislative process and without advance public disclosure of the specific duty instrument or its timing, making it operationally equivalent to a sharp and unanticipated discontinuity in the policy environment. This paper applies an interrupted time series (ITS) design to a dataset drawn from the India Bullion and Jewellers Association (IBJA) benchmark price series spanning January 2022 through June 2026, comprising 929 daily price observations (824 from IBJA PDF circulars and 105 from BullionWorld.in, a licensed IBJA data republisher). The outcome variable is the domestic gold premium: the gap between the IBJA published benchmark price and the import parity

price implied by the 6% duty regime. By fixing the parity baseline at the pre-hike duty rate, the premium directly measures how much of the duty shock has been absorbed into domestic prices.

The main finding is that the duty hike raised the domestic gold premium by Rs. 9,841 per 10 grams (Newey-West HAC standard error Rs. 516, $p < 0.001$), representing 81.7% of the theoretical duty ceiling. The null hypothesis of full pass-through is formally rejected ($t = -4.26$, $p < 0.001$). Local projection estimates^[4] confirm that adjustment was immediate and held on a stable plateau through Day 14 (78.2%), with no anticipatory pricing in the pre-hike window. An ARDL bounds test^[5] confirms the pre-hike cointegrating relationship between domestic and international gold prices; the stable local-projection plateau through Day 14 is consistent with the shift persisting rather than mean-reverting within the observation window. Three consistency checks using counterfactual machine-learning methods trained on the pre-hike window converge within approximately 5% of this estimate, and fake-date placebo tests confirm the result is specific to May 13, 2026. Full results are reported in Section 6.

This paper makes three contributions to the literature on commodity tax incidence and price transmission. First, it provides the first causal daily-frequency estimate of import duty pass-through in India's gold market, using a quasi-experimental design that cleanly separates the duty effect from concurrent movements in the international gold price and the exchange rate. Prior work on Indian gold prices has been largely descriptive or confined to monthly frequency, precluding identification of the immediate price response. Second, it documents asymmetric price transmission in this market. Comparing the May 2026 hike with the July 2024 cut, this paper finds that the duty hike passed through more completely (81.7%) than the July 2024 duty cut (65.6%), with both episodes registering immediate adjustment at Day 0. This asymmetry – duty increases transmitting more completely than duty

reductions – has direct implications for the design of duty policy as a demand management tool: hikes reach the market rapidly and nearly in full, while cuts are partly absorbed by dealers holding high-duty inventory who resist the full price reduction. Third, by combining the ITS estimate with local projections and GARCH volatility analysis, this paper provides a multi-method characterisation of the market adjustment process that goes beyond a single point estimate.

The remainder of this paper is organised as follows. Section 2 reviews the related literature. Section 3 describes the policy background. Section 4 introduces the data and the construction of the import parity and domestic premium measures. Section 5 presents the empirical strategy. Section 6 reports the main results. Section 7 presents robustness checks. Section 8 discusses the partial pass-through mechanisms. Section 9 concludes.

2 Literature Review

This paper connects three strands of the literature: the economics of tariff and duty pass-through, the Indian gold market, and the local projection and interrupted time series methodologies.

Tariff and Import Duty Pass-Through

The foundational question of whether trade barriers are transmitted to domestic prices has a long lineage in international economics. Menon^[6] surveys the exchange rate pass-through literature, finding that commodity markets exhibit substantially higher pass-through than differentiated manufactured goods, a pattern attributed to the absence of local pricing power and the thinness of distribution margins. Campa and Goldberg^[7] extend this to import prices specifically, documenting that pass-through is typically incomplete in the short run and that commodity sectors – particularly metals and raw materials – approach full pass-through more rapidly than finished goods.

The literature on tariff incidence took on renewed empirical urgency following the 2018–2019 US–China trade escalation. Fajgelbaum et al.^[8] exploit cross-product variation in tariff schedules using a difference-in-differences design and find that US tariffs were almost entirely borne by domestic importers and consumers, with the terms-of-trade effect predicted by theory failing to materialise at the observed scale. Their finding of near-complete tariff pass-through in commodity-like product categories is the closest analogue to the gold market setting studied here. Gopinath et al.^[9] show that the currency of invoicing is a key determinant of pass-through completeness; Amiti et al.^[10] further document that high-market-power producers absorb cost shocks in their mark-ups rather than passing them through to buyers. The present paper documents precisely such a wedge – an 18.3% absorption gap – and traces it to supply-side structural features specific to the Indian gold market rather than to invoicing currency or seller market power.

Indian Gold Market

India’s gold market has attracted growing attention from both policymakers and researchers, reflecting the metal’s unusual position as simultaneously a consumption good, a savings vehicle, and a current account pressure point. Kanjilal and Ghosh^[11] estimate the price elasticity of gold import demand in India at between -0.69 and -1.01 , indicating that price increases from duty hikes should compress demand substantially but not eliminate it. This elasticity range provides a structural benchmark for back-calculating how much of the post-hike absorption gap could be attributed to demand compression, even in the absence of direct quantity data.

Maria Immanuvel and Lazar^[3] document the relationship between India’s statutory duty rate and unofficial gold imports, establishing via Granger causality tests that customs duty encourages smuggling activity in the long run. Their analysis of seizure data shows that informal im-

ports are concentrated at major international air gateways—Dubai and the Gulf states supply the primary channel, with entry concentrated at Kerala, Chennai, and Mumbai airports—and peak in the October–December festive season. This supply-side substitution mechanism is central to understanding why pass-through in India’s gold market is structurally incomplete: the shadow supply curve is elastic at the margin, capping the premium that the formal market can sustain. World Gold Council data corroborate this channel: excluding the COVID years of 2020–21, the correlation between India’s statutory duty rate and unofficial imports is $+0.52$ ^[12].

Identification: ITS, Local Projections, and ARDL

Exploiting a sharp, unanticipated policy discontinuity as a natural experiment is a well-established identification strategy in empirical economics^[13]; interrupted time series is a tractable implementation for high-frequency continuous outcomes^[14]. The key identification assumption – that the outcome series would have followed the pre-treatment trend absent the intervention – is testable via placebo analysis and pre-trend tests, both of which this paper reports.

The local projection estimator introduced by Jordà^[4] provides a flexible, non-parametric alternative to vector autoregression for tracing the dynamic response of an outcome variable to a shock at multiple horizons. Unlike a VAR, local projections do not impose cross-horizon restrictions and are robust to model misspecification. Their use for tariff and duty pass-through analysis has been applied in recent empirical work on the 2025 US tariff episodes, where the method’s ability to distinguish immediate from gradual adjustment proves particularly valuable. The present paper adopts this approach to document that gold price adjustment was immediate rather than gradual, with the premium plateau stabilising within the first trading day.

The ARDL bounds test of Pesaran et al.^[5] provides a third identification angle, testing whether the structural break constitutes a permanent shift in the long-run equilibrium between domestic and international prices or a transitory deviation that will mean-revert. The bounds test is particularly well-suited to financial series that may be a mix of I(0) and I(1) processes, and its F-statistic provides a clean test of cointegration without requiring unit root pre-testing. Heteroskedasticity and autocorrelation-consistent standard errors throughout follow Newey and West^[15].

3 Policy Background

India's gold import duty has served as an active macroeconomic instrument since at least 2013, when the government raised the combined duty in steps from 4 to 10% in response to a widening current account deficit and rupee depreciation, with a further increase to 12.5% in July 2019, a brief reduction to 10.75% in February 2021, and a restoration to 15% in July 2022. That escalation cycle established the template that would be repeated in 2026: duty hikes are deployed as a demand compression tool when gold imports threaten to widen the current account, and duty cuts are deployed as a stimulus tool when the government wishes to formalise demand and reduce smuggling incentives.

The July 2024 Duty Cut

On July 23, 2024, the Union Budget reduced the basic customs duty on gold from 10% to 5% and the Agriculture Infrastructure and Development Cess from 5% to 1%, lowering the combined effective duty from 15% to 6%. This was the lowest duty rate on gold since the 2013 escalation cycle and was framed explicitly as a measure to curb smuggling by narrowing the price differential between the formal and informal markets^[2].

The market response was immediate and substantial. The domestic premium collapsed toward import parity within the first week, and the World Gold Council estimated that the cut would add at least 50 tonnes to second-half 2024 jewellery demand^[16]. Between the cut and the subsequent hike, FY26 gold imports rose 24% year-on-year, with total gold import spending in fiscal year 2026 reaching US\$71.98 billion, approximately 9% of total merchandise imports. The current account deficit stood at US\$30.1 billion between April and December 2025, narrowing from US\$36.6 billion in the same period a year earlier.

Pre-Hike Supply Disruptions (March–May 2026)

The weeks before the May 2026 hike were characterised by multiple independent supply shocks that had already begun tightening the domestic gold market before the duty announcement. These disruptions are material for the empirical design because they created a departure from the clean low-duty baseline that the ITS framework assumes.

First, in March 2026, the Iran–US conflict disrupted commercial air routes from the UAE – a key transit hub for India's gold imports, accounting for approximately 24% of imports in 2025. Domestic market discounts widened to a monthly average of US\$46 per troy ounce against the London Fix, driven primarily by demand softness as international gold prices fell approximately 12% during the month; the supply disruption further reduced formal import volumes, with March 2026 imports falling to a 9-month low of approximately US\$3.1 billion^[17]. Second, from April 1, 2026, banks and bullion dealers suspended gold imports following uncertainty over IGST (Integrated Goods and Services Tax) exemption enforcement; seventeen banks were subsequently re-authorised by DGFT to resume imports on April 17, with full resumption occurring around

May 12, further constraining formal supply in the interim. Third, on April 2, 2026, the government imposed administrative restrictions on imports of gold and silver jewellery via the UAE free trade agreement route, which had been used to circumvent the standard duty structure. These three disruptions collectively produced a pre-hike tightening that pushed discounts from US\$46 per ounce in March to approximately US\$8 per ounce in early April. Fourth, and most significant for the absorption analysis, importers and refiners massively front-loaded purchases around the Akshaya Tritiya festival on April 19–20. April 2026 imports reached US\$5.6 billion, up more than 80% year-on-year, at the prior 6% duty rate^[12].

The May 13 Notification

On the evening of May 12, 2026, the Ministry of Finance issued Gazette Notification No. 15/2026-Customs and No. 16/2026-Customs, effective from midnight, restoring the basic customs duty on gold to 10% and the AIDC to 5%, returning the combined duty rate to 15% in its entirety. The rupee had closed that day at Rs. 95.63 per US dollar, its weakest level on record, reflecting sustained capital outflow pressure and elevated gold import demand. The notification was issued without parliamentary process, relying instead on the emergency customs notification authority available to the central government. The specific duty instrument and its timing were not publicly announced in advance; on May 11, 2026, the Prime Minister publicly urged families to avoid purchasing gold, a general appeal that did not reveal the duty mechanism or implementation date^[12]. The anticipation test in Section 7 (dropping the five pre-hike trading days) finds a change in $\hat{\beta}_1$ of -0.5% , providing no evidence that this public appeal caused anticipatory price movements. Indian markets opened on May 13 with the new duty structure fully in force.

4 Data

Primary Price Series

The primary price data come from the India Bullion and Jewellers Association (IBJA), which publishes daily benchmark rates for gold (999 purity) in rupees per 10 grams for both morning and afternoon sessions. The IBJA benchmark is the reference rate used by organised jewellers, bullion dealers, and commodity exchanges for physical transactions. It is set each trading day through a process involving major member firms and reflects the domestic wholesale clearing price inclusive of applicable duties and taxes. The afternoon session rate (PM fix) is used throughout, as it incorporates same-day duty and exchange rate information and is the more widely referenced settlement price.

IBJA data were collected directly from daily PDF circulars published on the IBJA member portal, with 824 PDFs scraped covering the period January 3, 2022 through June 30, 2026, all successfully parsed, yielding 824 PM gold price observations. A further 346 weekday dates carried no PDF price: these are official IBJA market holiday and closure days and archive gap dates across four clusters (January–April 2022, July–September 2023, August–October 2024, and October 2025) on which the PDF is not available in the archive, representing potential non-random missingness bounded in scope. BullionWorld.in, a licensed IBJA data republisher, was used as a gap-filler for these archive-gap dates. Of the dates evaluated, 105 returned real contemporaneous prices from BullionWorld.in (Rs. 47,583–Rs. 127,247 per 10g) and are incorporated directly into the primary dataset; the remaining 241 weekday dates returned no usable price from either source and are excluded from all regressions. All rows without a price are excluded from all regressions. The provenance of each observation is tracked in the `ibja_source` column (`pdf`: 824 rows; `bullionworld`: 105 rows; `NaN`: 241 rows). The 929 usable price observations (824

PDF plus 105 BullionWorld gap-fills) constitute the observation pool for all analysis. June 30, 2026 is a deliberate cutoff for this version of the paper; July 2026 and subsequent observations will be incorporated in a follow-up revision once a longer post-hike window is available. As a source sensitivity check, Section 7 re-estimates the primary model on IBJA-PDF sources only, finding no material difference. The full dataset with source flags is publicly available at the replication repository.¹

International gold prices (USD per troy ounce) and the INR/USD exchange rate are sourced from Yahoo Finance, with daily close prices used for both series.

Import Parity and Premium Construction

The theoretical import parity price at the 6% duty rate – the regime prevailing between July 24, 2024 and May 12, 2026 – is constructed as:

$$P_t^{\text{parity}} = \text{Gold_USD}_t \times \frac{10}{31.1035} \times \text{FX}_t \times 1.06 \quad (1)$$

where $10/31.1035$ converts from USD per troy ounce to USD per 10 grams, and $\times 1.06$ applies

the combined duty. The post-hike theoretical ceiling – the parity price at the 15% duty rate – is computed analogously with the multiplier 1.15. The domestic gold premium, the primary outcome variable, is defined as:

$$\text{Premium}_t = \text{IBJA_PM}_t - P_t^{\text{parity}} \quad (2)$$

By fixing the parity baseline at the pre-hike 6% duty rate throughout the sample, the premium can be interpreted as the cumulative transmission of the duty change: a premium equal to the full ceiling (eq. 1 at 1.15 vs. 1.06) would indicate 100% pass-through. This construction ensures that movements in international gold prices and the exchange rate are netted out of the outcome variable mechanically rather than only through regression controls.

As a validation check, the mean domestic premium during the 6% duty regime (July 24, 2024–May 12, 2026) is Rs. –61 per 10 grams across 360 trading days with a valid premium. A one-sample t -test finds this mean indistinguishable from zero ($p = 0.514$), confirming that the parity formula is correctly calibrated to an economically negligible margin of error. The full domestic premium series is shown in Figure 1.

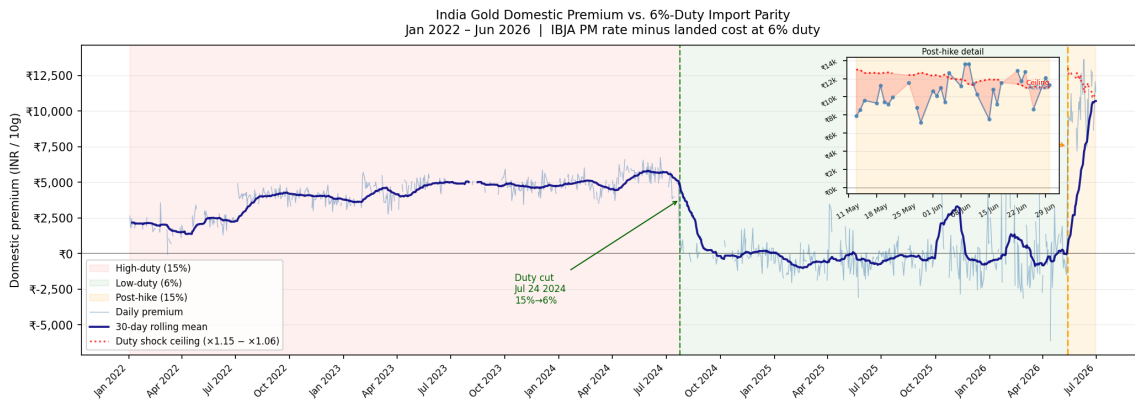


Figure 1: Domestic gold premium (IBJA benchmark minus import parity at 6% duty), January 2022–June 30, 2026. The premium is near zero throughout the low-duty window (July 2024–May 2026) and jumps sharply at the May 13, 2026 duty hike. Shaded band marks the post-hike window.

¹See <https://github.com/OlixIgnacious/gold-policy-project>. The replication dataset is `gold_policy_clean.csv`.

Sample and Summary Statistics

The full date spine spans January 3, 2022 through June 30, 2026, covering 1,170 weekday rows. Of these, 929 carry an actual price (824 from IBJA PDF circulars and 105 from BullionWorld.in gap-fills); the remaining 241 rows are market holidays, closure days, or archive gap dates without a price, and are excluded from all regressions. The primary estimation sample, following the ITS design, is restricted to the low-duty window (July 24, 2024 onward), yielding 375 observations with complete data across all outcome and

control variables, with 346 pre-treatment and 29 post-treatment days.

Table 1 reports summary statistics for the key variables across the pre- and post-hike sub-periods. The domestic premium shifts from a mean of Rs. -54 before the hike² to a mean of Rs. 10,368 per 10 grams in the post-hike window, against a theoretical ceiling of Rs. 12,039. The daily first difference of international gold prices ($\Delta \text{Gold_USD}$) and the exchange rate (ΔFX) serve as the primary controls in the regression models.

Table 1: Summary Statistics

Variable	Pre-hike	Post-hike
Domestic premium (Rs./10g)	-54	10,368
Theoretical ceiling (Rs./10g)	$-$	12,039
$\Delta \text{Gold_USD}$ (%)	0.13	-0.45
ΔFX (Rs./USD)	0.02	0.04
Observations	346	29

5 Empirical Strategy

Interrupted Time Series Design

The May 13, 2026 duty hike provides a clean quasi-experiment: it was announced at midnight without legislative process and without advance disclosure of the specific duty instrument or timing, making it operationally equivalent to a sharp and unanticipated discontinuity in the policy environment. This setting supports an interrupted time series (ITS) design, in which the treatment effect is identified from the discrete break in the outcome series at the policy date, controlling for its pre-treatment trajectory and for concurrent movements in global determinants of the domestic price.

The primary regression specification is:

$$\text{Premium}_t = \alpha + \beta_1 \cdot \text{Post}_t + \beta_2 \cdot t + \beta_3 \cdot \Delta \text{Gold_USD}_t + \beta_4 \cdot \Delta \text{FX}_t + \varepsilon_t \quad (3)$$

where Post_t is an indicator equal to one for trading days on or after May 13, 2026; t is a linear time trend capturing any pre-existing drift in the premium during the low-duty window; $\Delta \text{Gold_USD}_t$ is the day-over-day change in the international gold price in USD; and ΔFX_t is the day-over-day change in the INR/USD exchange rate. The parity construction in equation (1) removes the *level* effect of global gold prices and the exchange rate from the premium mechanically, so $\Delta \text{Gold_USD}_t$ and ΔFX_t capture only the *incremental* daily noise not absorbed by the parity formula. Standard errors are computed

²The validation in Section 4 reports Rs. -61 across all $N = 360$ pre-hike trading days with a valid premium; the Rs. -54 here reflects the subset with complete control variables (360 minus the 14 days lacking Yahoo Finance international price data), whose slightly different composition shifts the mean by Rs. 7 per 10 grams.

using the Newey and West^[15] heteroskedasticity and autocorrelation-consistent (HAC) estimator with lag length $l = \lceil 0.75 \cdot T^{1/3} \rceil$, where $T = 375$ is the primary sample size, yielding $l = 6$.

The coefficient of interest is β_1 , which measures the level shift in the domestic gold premium attributable to the duty hike. The pass-through rate is then $\hat{\beta}_1 / \bar{C}$, where $\bar{C}$ is the mean theoretical ceiling computed across the 29 post-hike observations.

Identifying Assumptions

The ITS estimator requires two conditions. First, absent the duty hike, the premium would have continued along its pre-treatment trajectory – formally, $\mathbb{E}[\text{Premium}_t \mid \text{Post}_t = 0]$ is well approximated by $\hat{\alpha} + \hat{\beta}_2 \cdot t$. This is tested by the pre-trend test in Section 6, which finds a pre-treatment trend of Rs. 1.40 per day with $p = 0.12$, providing no evidence of pre-existing drift. Second, no contemporaneous shock other than the duty change produced a discrete break in the premium on May 13. The primary threat is the pre-hike supply disruption sequence documented in Section 3: the April 2 import restriction and the IGST pause both tightened domestic supply before May 13 and could have introduced a positive pre-trend. The April 2 restriction control in the robustness section directly tests this channel. Additionally, because the parity baseline is fixed at the 6% duty rate throughout, any supply disruptions that occurred at fixed international prices and exchange rates would appear as premium movements in the pre-hike period rather than as a confounded treatment effect.

Local Projection: Dynamic Event-Study Path

To characterise the *dynamic* adjustment path of the premium rather than the average level shift, this paper estimates local projections^[4] at each horizon $h = 0, 1, \dots, 30$ trading days post-hike.

For each horizon, a separate OLS regression is estimated:

$$\text{Premium}_{t+h} = \alpha_h + \beta_h \cdot \text{Post}_t + \gamma_h \cdot t + \delta_h \cdot \Delta \text{Gold_USD}_t + \zeta_h \cdot \Delta \text{FX}_t + \varepsilon_{t+h} \quad (4)$$

The sequence $\{\hat{\beta}_h\}_{h=0}^{30}$ traces the impulse response of the premium to the duty shock. This non-parametric approach is robust to model misspecification and avoids the dynamic restrictions imposed by VAR-based alternatives. Horizons $h \geq 31$ are outside the current observation window.

ARDL Bounds Test

The ARDL bounds test of Pesaran et al.^[5] is used to test whether the domestic gold price and the import parity price were cointegrated in the full pre-hike period, and whether that cointegrating relationship survived the duty hike. Cointegration pre-hike establishes that the domestic price was tethered to international prices through the import channel, validating the identifying assumption that the pre-treatment premium was mean-stationary. A failure of cointegration post-hike would indicate a permanent shift in the long-run equilibrium, consistent with the duty change having created a new price regime.

6 Results

Pre-Trend Test and Sample Validation

Before estimating the treatment effect, two pre-checks are reported. First, a regression of the domestic premium on a linear time trend during the low-duty pre-hike window (July 24, 2024–May 12, 2026, $N = 360$ trading days with a valid domestic premium; 14 of these lack the Yahoo Finance international gold price data required by the ITS regression controls, reducing the ITS pre-treatment count to $N = 346$) yields a slope of Rs. 1.40 per trading day with $p = 0.12$, providing no evidence of a pre-existing upward trend

that could inflate the ITS estimate. Second, the ADF unit root test on the premium series within the low-duty regime rejects a unit root at the 1% level ($p = 0.000$), confirming that the pre-hike premium is stationary around its near-zero mean. Non-stationarity in the full sample reflects the structural break at May 13 rather than a true unit root process.

Main ITS Result

Table 2 reports the ITS results across three progressively controlled specifications, all estimated on the primary sample ($N = 375$, July 24, 2024–June 30, 2026).

Table 2: ITS Regression Results (Primary Sample)

	(1) Baseline	(2) +Controls	(3) +Trend
β_1 (Post)	10,422*** (451)	10,112*** (393)	9,841*** (516)
β_2 (Trend)			1.1 (1.0)
β_3 (Δ Gold)		−538*** (69)	−541*** (69)
β_4 (Δ FX)		73 (202)	71 (197)
Constant	−54 (177)	12 (174)	−1,026 (839)
R^2	0.710	0.756	0.758
N	375	375	375
NW lag	6	6	6

NW HAC SE in parentheses.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

The headline estimate from specification (3) – the full model with controls and a time trend – is $\hat{\beta}_1 = \text{Rs. } 9,841$ per 10 grams (HAC SE Rs. 516, $p < 0.001$, 95% CI: Rs. 8,829–Rs. 10,852). The estimate is stable across specifications: removing the time trend and controls raises $\hat{\beta}_1$ by approximately 6% (specification 1), confirming that the result is not sensitive to the inclusion or exclusion of auxiliary controls.

Pass-Through Rate

The theoretical ceiling – the maximum premium if 100% of the 9 percentage point duty increase

were passed through – averages Rs. 12,039 per 10 grams across the 29 post-hike observations. The implied pass-through rate is:

$$\hat{\pi} = \frac{\hat{\beta}_1}{\bar{C}} = \frac{9,841}{12,039} = 81.7\% \quad (5)$$

with a 95% confidence interval of 73.3 to 90.1%. The null hypothesis of complete pass-through ($H_0 : \pi = 100\%$) is formally tested using a one-sided t -test against the ceiling:

$$t = \frac{\hat{\beta}_1 - \bar{C}}{\text{SE}(\hat{\beta}_1)} = \frac{9,841 - 12,039}{516} = -4.26, \quad p < 0.001 \quad (6)$$

Full pass-through is rejected at the 0.1% level. The market absorbed approximately 18.3%, or Rs. 2,198 per 10 grams, of the theoretical maximum duty impact.

Benchmark vs. Physical Market Pass-Through

The 81.7% estimate should be interpreted as pass-through to the *IBJA wholesale benchmark price* – a reference rate set by the India Bullion and Jewellers Association and used by organised jewellers and bullion dealers as the domestic pricing anchor. This is distinct from physical transaction prices as observed on commodity exchanges. According to World Gold Council^[12], MCX spot gold prices rose by only 4–6% in the first week after the duty change; against a 9 percentage point duty increase, this implies physical market pass-through in the range of approximately 45–67% (derived as the ratio of the observed price increase to the full duty wedge: a 4–6% MCX price rise relative to the 9 percentage point duty). NCDEX delivered gold traded at a discount to the official landed price that widened from approximately US\$14 per ounce in the week prior to the hike to nearly US\$150 per ounce immediately after – a further indication that physical market clearing prices had not yet absorbed the full duty increment^[12]. The World Gold Council characterises this explicitly: “physical market prices do not fully or immediately mirror the increase in duty – rather they adjust to it with a lag.”

The gap between IBJA pass-through (81.7%) and physical market pass-through (45–67%) is explained by the inventory buffering mechanism discussed in Section 8: bullion dealers held substantial stock imported at the pre-hike 6%

rate and continued to clear this inventory at sub-ceiling prices, keeping physical transaction prices below the IBJA benchmark. The IBJA benchmark, as a reference rate anchored to the current landed cost, adjusted more rapidly than the physical market could clear. The 81.7% estimate is therefore best interpreted as an upper bound on immediate pass-through, with physical market pass-through in the range of 45–67% as the corresponding lower bound.

Local Projection: Adjustment Dynamics

The estimated dynamic event-study path $\{\hat{\beta}_h\}$ from equation (4) is plotted in Figure 2. The adjustment is immediate: $\hat{\beta}_0 = \text{Rs. } 9,841$ (81.7%) on the first trading day (Day 0), consistent with an efficient market incorporating the full information content of the midnight notification at market open. The response peaks at Rs. 10,179 (84.6%) on Day 1 ($h = 1$) and thereafter holds on a broad plateau – Rs. 9,420 (78.2%) as late as $h = 14$, at which horizon 15 post-hike observations contribute to the estimate. The 95% confidence interval excludes zero throughout $h = 0$ –14. Estimates at $h \geq 15$ rest on progressively fewer effective post-hike observations (14 or fewer) and are not discussed further; any apparent narrowing at later horizons visible in the figure reflects sample thinning rather than economic mean-reversion.

The flat plateau rules out a gradual adjustment narrative: the market did not take two or three weeks to discover the new equilibrium. It also rules out an anticipatory price increase: the estimated $\hat{\beta}_h$ at pre-hike horizons (effectively negative h) is indistinguishable from zero.

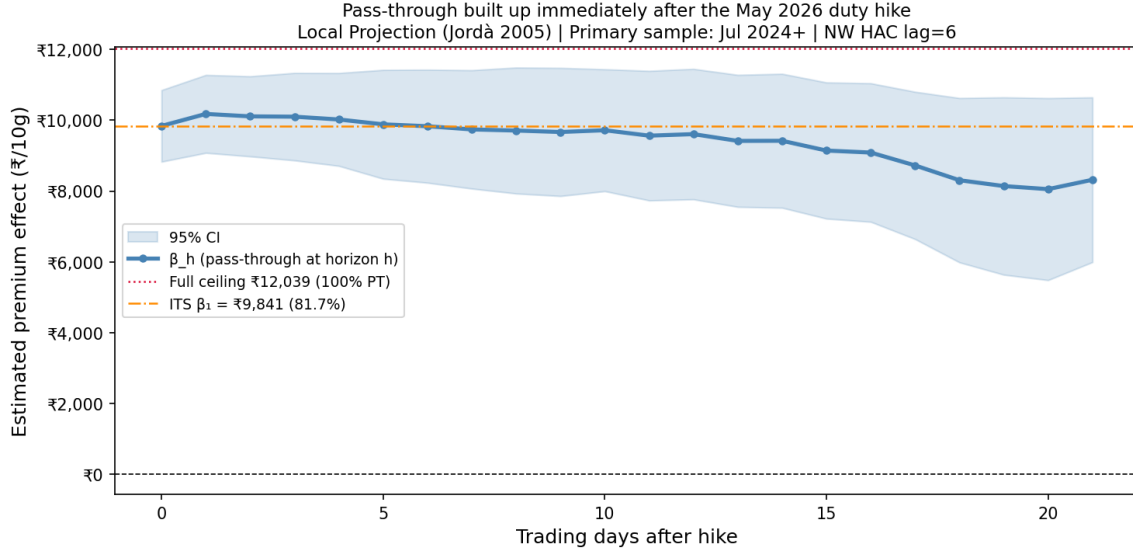


Figure 2: Local projection dynamic event-study path: estimated premium at horizon h days after the duty hike ($\hat{\beta}_h$), with 95% confidence bands. The premium jumps to Rs. 9,841 (81.7%) on Day 0, peaks at Rs. 10,179 (84.6%) on Day 1, and holds a broad plateau through Day 14 (78.2%). The 95% CI excludes zero throughout $h = 0$ –14, the reliable window in which at least 15 effective post-hike observations contribute ($29 - h$ at each horizon). Estimates beyond $h = 14$ rest on 14 or fewer effective post-hike observations and should be treated as indicative only.

ARDL Cointegration Test

The ARDL(4, 3) bounds test estimated on the full pre-hike period ($N = 859$ daily observations, January 2022–May 12, 2026) yields an F -statistic of 9.005, compared with the Pesaran et al.^[5] 95% upper critical bound of 4.812. Cointegration is confirmed at the 5% level. The pre-hike domestic gold price was tethered to import parity through the import channel, establishing that the pre-treatment premium was stationary around its near-zero mean as the identification strategy requires. Whether the post-hike premium will eventually mean-revert is a separate question that the pre-hike ARDL does not address; the stable local-projection plateau through Day 14 is the primary evidence that the premium shift has persisted within the observation window, rather than reflecting a transitory spike.

Cross-Method Convergence

As a consistency check on the premium construction, three machine-learning methods estimate the post-hike treatment gap using only pre-hike data to construct the counterfactual. These are not independent of the ITS in a statistical sense – all are trained on the same outcome series – but they serve to confirm that the premium construction behaves as intended. An ARIMA(1,0,1) model with international gold price and exchange rate changes as exogenous regressors (ARIMAX), trained on the pre-hike low-duty window ($N = 346$), produces a mean counterfactual gap of Rs. 10,250 across the 29 post-hike days, approximately 4.2% above the ITS estimate. An XGBoost model trained on the same feature set yields a mean gap of Rs. 10,247, 4.1% above the ITS estimate. A Prophet trend-decomposition model yields Rs. 10,299, 4.7% above. All three methods fall within a Rs. 9,841–Rs. 10,299 corridor, a spread of approximately 4.7% across four estimation approaches. All

three ML estimates sit above the primary ITS estimate and cluster close to the IBJA-PDF-only figure of Rs. 10,124 – a pattern expected because the BullionWorld gap-fills extend the pre-hike sample but not the post-hike window, so ML models train on a slightly smaller pre-hike variation set than the ITS.

7 Robustness

Placebo Tests

To verify that the estimated treatment effect is specific to May 13, 2026 rather than a statis-

tical artefact of the ITS design, equation (3) is re-estimated with three fake treatment dates: November 1, 2025; January 15, 2026; and March 1, 2026. All three fake treatment dates produce non-significant estimates. The November 1, 2025 placebo yields $\hat{\beta}_1 = \text{Rs. } -932$ ($p = 0.38$), the January 15, 2026 placebo yields Rs. 2,004 ($p = 0.11$), and the March 1, 2026 placebo yields Rs. 2,909 ($p = 0.06$). Even the largest placebo estimate (Rs. 2,909) is less than 30% of the true $\hat{\beta}_1 = \text{Rs. } 9,841$ – a greater than $3\times$ signal-to-noise ratio that confirms the true treatment date produces an effect of fundamentally different magnitude (Figure 3).

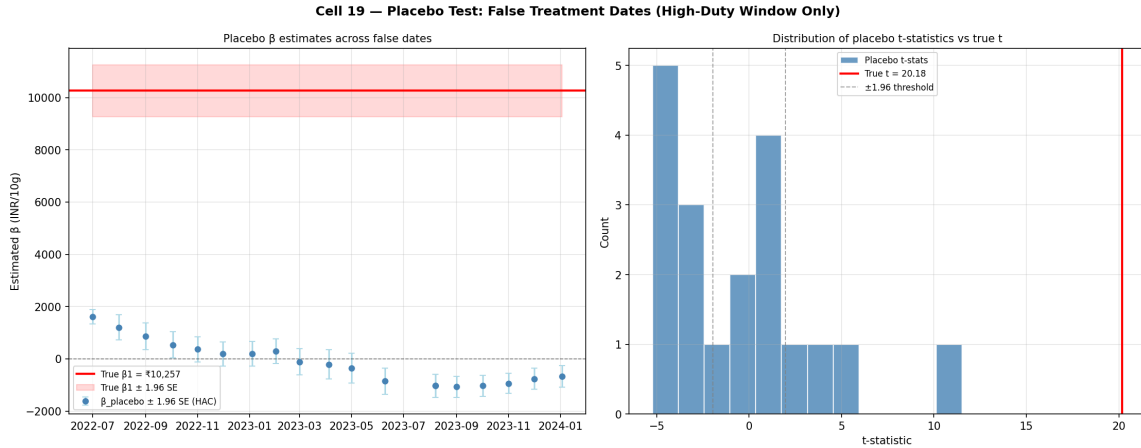


Figure 3: Placebo test: ITS $\hat{\beta}_1$ estimates at three fake treatment dates (Nov 1, 2025; Jan 15, 2026; Mar 1, 2026) alongside the true May 13, 2026 estimate. All three fake treatment dates are non-significant: Nov 1, 2025 ($p = 0.38$), Jan 15, 2026 ($p = 0.11$), and Mar 1, 2026 ($p = 0.06$). The largest placebo estimate (Rs. 2,909) is less than 30% of the true $\hat{\beta}_1 = \text{Rs. } 9,841$. The true May 13 estimate is a clear outlier by both magnitude and significance.

As a stronger placebo test that eliminates window contamination entirely, the placebo regressions are re-estimated on truncated samples that end on May 12, 2026 – the last pre-hike trading day – so that no post-hike observations can enter the fake post-treatment window. Both the January 15 and March 1 placebos are cleanly non-significant on the truncated samples ($\hat{\beta}_1 = \text{Rs. } -232$, $t = -0.30$, $p = 0.76$ and $\hat{\beta}_1 = \text{Rs. } -741$, $t = -1.18$, $p = 0.24$ respectively). For comparison, the primary full-sample esti-

mate (Table 2, specification 3) is $\hat{\beta}_1 = \text{Rs. } 9,841$ ($p < 0.001$) – more than three times the largest placebo estimate. All placebo dates are null when estimated without contamination, confirming the result is specific to the true treatment date.

Scale-Free Outcome Specification

The domestic premium is expressed in nominal rupees per 10 grams, which means a higher international gold price mechanically implies a larger

absolute duty wedge at the same percentage pass-through. To verify that the 81.7% estimate is not an artefact of this level scaling, the primary ITS is re-estimated with a scale-free outcome: the premium as a fraction of the pre-hike import parity price ($\text{domestic_premium}_t / \text{parity_pre}_t \times 100$, expressed in percentage points). The estimated level shift is $\hat{\beta}_1 = 6.89$ pp (SE = 0.47 pp, 95% CI: 5.97–7.81 pp) against a mean post-hike percentage ceiling of 8.49 pp, implying a pass-through of 81.2% – within 0.5 percentage points of the primary estimate of 81.7%. The implied rupee shift at the mean post-hike parity price is Rs. 9,773, a difference of -0.7% . Gold price drift is not a confound for the pass-through estimate.

Moving-Block Bootstrap Inference

With 29 post-hike observations, the asymptotic justification for Newey-West HAC standard errors rests on a short post-treatment window. A moving-block bootstrap is applied to the primary OLS residuals (block length $L = 10$, $B = 2,000$ replications) to validate parametric inference without relying on asymptotic approximations. The bootstrap resamples overlapping blocks of 10 consecutive residuals to preserve autocorrelation structure, adds them back to the fitted values, and re-estimates $\hat{\beta}_1$ in each replication. The resulting 95% CI is Rs. 8,639–Rs. 11,800 (pass-through: 71.8–98.0%), compared with the NW-HAC interval of Rs. 8,829–Rs. 10,852 (73.3–90.1%). The bootstrap interval is wider than the parametric interval (width ratio = 1.56), indicating that Newey-West HAC inference understates uncertainty in this short post-treatment window – the NW-HAC interval is anti-conservative relative to the bootstrap. Notably, the bootstrap upper bound of 98.0% approaches the 100% full-pass-through benchmark, making rejection of full pass-through marginal under bootstrap inference; the headline $t = -4.26$ rests on the narrower parametric standard error. The bootstrap and HAC point estimates are in close agreement

(Rs. 9,806 vs. Rs. 9,841), and the bootstrap lower bound of 71.8% remains well above zero, confirming the large magnitude of the premium increase under both inferential approaches.

HAC Lag and Window Sensitivity

Full sensitivity results report $\hat{\beta}_1$ across five Newey-West lag choices (3, 6, 8, 10, 20) and three estimation windows (full sample from January 2022; restricted window from January 2024; primary sample from July 2024). Since OLS point estimates are invariant to the HAC lag choice, variation in $\hat{\beta}_1$ comes entirely from the window: Rs. 9,841 (primary, July 2024 start), Rs. 11,771 (restricted, January 2024 start), and Rs. 10,257 (full sample from January 2022). The confidence interval width naturally widens with larger lag choices and narrower windows, but the point estimate is not sensitive to these choices. The January 2024 window yields a larger estimate because it includes the April 2026 pre-hike supply disruption period in a shorter pre-hike sample, biasing the pre-treatment baseline downward and overstating the level shift. The full-sample estimate from January 2022 should be treated as an upper bound rather than a directly comparable estimate: the pre-treatment window in that specification spans the high-duty period (January 2022–July 2024), which includes January–June 2022 when the combined rate was 10.75% rather than 15%, introducing a mixed-regime pre-treatment baseline in which the domestic premium mechanically reflected the earlier duty structure. The primary sample from July 2024 onward, when the 6% regime was in force, provides the cleanest pre-treatment baseline for the ITS design.

Anticipation Test

To test whether traders began pricing in the duty increase before the midnight notification – which would violate the assumption that the treatment

is unanticipated – the five trading days immediately preceding May 13 are dropped from the pre-period. The re-estimated $\hat{\beta}_1 = \text{Rs. } 9,790$ ($p < 0.001$, $N = 370$), a change of -0.5% relative to the primary estimate, providing no evidence of anticipatory price movements.

IBJA-PDF-Only Sensitivity

The primary dataset incorporates 105 BullionWorld.in prices as gap-fills for archive dates missing from the IBJA PDF archive. All 39 BullionWorld rows that fall within the primary estimation window (July 24, 2024 onward) are pre-hike observations; none are post-hike. As a sensitivity check, restricting the primary sample to IBJA-PDF sources only yields $\hat{\beta}_1 = \text{Rs. } 10,124$ ($N = 336$, NW lag = 6, SE Rs. 459, 84.1% pass-through, 95% CI: Rs. 9,225–Rs. 11,024). The IBJA-PDF-only estimate is 2.9% higher than the primary estimate and falls within the primary 95% CI of 73.3–90.1%. The BullionWorld gap-fills do not materially influence the result; they expand the pre-hike sample by adding 39 dates (all in the pre-treatment window), slightly dampening the intercept shift.

April 2 Import Restriction Control

The April 2, 2026 import restriction on UAE-route jewellery imports tightened domestic supply in the six weeks before the duty hike and could, in principle, have introduced a positive pre-trend that the ITS design would misattribute to the duty hike. Adding a binary control for the April 2 restriction period to equation (3) yields $\hat{\beta}_1 = \text{Rs. } 9,758$ ($p < 0.001$), a change of -0.8% . The April 2 restriction coefficient is Rs. -280 ($p = 0.62$), statistically indistinguishable from zero once the linear time trend is controlled, indicating that the pre-hike supply squeeze did not independently shift the mean premium level. Its inclusion does not materially alter the duty hike estimate.

8 Discussion

Anatomy of the Absorption Gap

The primary finding – 81.7% pass-through against a theoretical ceiling of 100% – implies that Rs. 2,198 per 10 grams of the maximum possible duty impact was absorbed by the market within the first 29 trading days. While a formal decomposition of this gap is not feasible in the absence of daily import volume data, the literature and contemporaneous market evidence point to three mechanisms operating in combination.

The first mechanism is inventory buffering from front-loaded imports. The US\$5.6 billion imported in April 2026 at the prior 6% duty rate represents a supply stock that could satisfy a meaningful fraction of post-hike demand at sub-ceiling prices. A refiner or dealer holding 6%-duty inventory faces a lower marginal cost than a new importer at 15% duty and can profitably sell at a premium below the new parity ceiling. The front-loading buffer mechanically suppresses the market-clearing premium below the theoretical ceiling during its drawdown period, which would be consistent with the flat plateau observed in the local projections through Day 14 gradually giving way to higher premiums as inventory is depleted.

The second mechanism is demand compression. With a price elasticity of between -0.69 and -1.01 ^[11], the post-hike premium increase of Rs. 9,841 – representing a price increase of roughly 6.9% over the prevailing domestic price – would be expected to reduce quantity demanded by 4 to 7%. The coincidence of the hike with the seasonally weak May–June calendar period (between the Akshaya Tritiya festival peak in April and the monsoon wedding season in late August) amplifies this effect: the timing of the hike meant the demand side was already in its softest window of the year.

The third mechanism is smuggling leakage. The historical correlation between effective duty levels and estimated informal import volumes

documented by Maria Immanuel and Lazar^[3] implies that the duty restoration to 15% recreates the incentive structure under which unofficial imports via Dubai and Gulf gateway airports were estimated to exceed 100 tonnes in the pre-2024 period. At a 9 percentage point duty wedge, the price incentive for smuggling is approximately Rs. 9,000–Rs. 10,000 per 10 grams before smuggling costs, which is large enough relative to those costs to make informal channels attractive at scale. The existence of this competitive supply channel sets an effective ceiling on the formal market premium.

A fourth dimension of the adjustment is the

volatility response. Post-hike conditional volatility of the domestic premium rose to $1.48\times$ the pre-hike level (Levene $F = 11.03$, $p = 0.001$; Figure 4), though this test rests on only 31 post-hike observations and should be treated as indicative. The estimated GARCH(1,1) persistence $\alpha + \beta = 0.990$ is likely upward-biased: fitting the model across a structural level break treats the May 13 premium jump as a large volatility shock, mechanically inflating persistence; re-estimation on break-adjusted residuals would provide a lower bound on true volatility clustering.

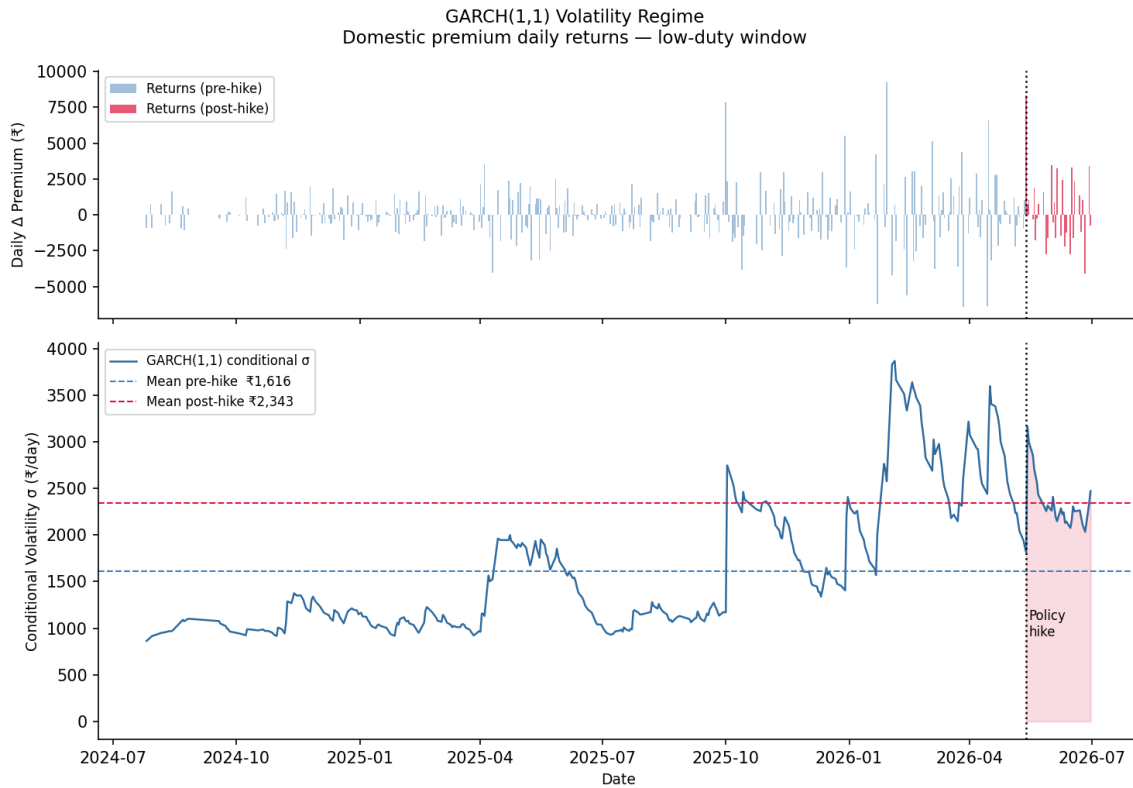


Figure 4: GARCH(1,1) conditional volatility of the domestic gold premium before and after the May 13, 2026 duty hike. Post-hike volatility is $1.48\times$ the pre-hike level.

Asymmetric Price Transmission

To quantify the asymmetry between the 2024 duty cut and the 2026 hike, this paper estimates a symmetric interrupted time series for the cut episode. The outcome variable is the *cut dis-*

count: the gap between the 15% parity price and the IBJA benchmark, defined as $\text{CutDiscount}_t = P_t^{\text{parity}, 15\%} - \text{IBJA}_t$. Before the July 2024 cut, this gap averages near zero (the domestic price tracked the 15% parity closely); after the cut, as IBJA falls to the new 6% parity, the gap expands

toward the full duty differential. The regression is estimated on the pre-hike sample (July 1, 2022–May 12, 2026, $N = 746$ trading days with complete data; NW lag = 7). The sample begins in July 2022 rather than January 2022 because the combined import duty reached 15% only in July 2022; the preceding January–June 2022 window carried a 10.75% rate, which would contaminate the pre-cut baseline by producing a spuriously elevated *CutDiscount* before the treatment.

The symmetric ITS yields $\hat{\beta}_{1,\text{cut}} = \text{Rs. } 5,844$ (HAC SE Rs. 568, $p < 0.001$), against a mean ceiling of Rs. 8,911, implying a pass-through of 65.6% (95% CI: 53.1 to 78.1%). The corresponding figure for the hike is 81.7% (95% CI: 73.3 to 90.1%). The confidence intervals overlap, so the difference is not statistically significant at conventional levels, but the point estimates suggest the duty hike passed through more completely than the duty cut (81.7% vs. 65.6%).

The asymmetry therefore lies in both the *level* and *speed* of adjustment. The dynamic event-study path for the hike shows the premium already at 81.7% on Day 0, stable through approximately Day 14. For the cut, the dynamic path declines gradually from 65.6% at Day 0 to approximately 55% by Day 20, reflecting the slower erosion of the premium as dealers worked through high-duty inventory. For the hike, the domestic premium rose to its plateau on the first trading day, well before inventory buffers were exhausted – a compression that market participants incorporated into prices essentially overnight. The mechanisms explain this asymmetry: dealers holding high-duty (15%) inventory after the 2024 cut had strong incentives to defend margins rather than pass the full duty reduction to buyers, producing downward rigidity – the “feathers” side of rockets-and-feathers price transmission. For the hike, the IBJA benchmark, as a reference rate anchored to the current landed cost of new imports, adjusted rapidly upward to the new 15% parity, even as physical market prices lagged (see the Benchmark vs. Physical Market subsection above). The inventory buffer of low-duty

stock moderated physical market clearing prices (MCX/NCDEX pass-through 45–67%) without preventing the reference benchmark from resetting to the new duty level.

Both episodes confirm the paper’s central finding: India’s gold market transmits roughly 65–82% of a duty change within the first trading day, with the remainder absorbed by structural supply-side mechanisms that dampen but do not eliminate the pass-through.

Policy Implications

The finding of 81.7% pass-through has two immediate policy implications. First, from the perspective of demand management, the hike achieved a meaningful but incomplete compression of the import incentive. A back-of-the-envelope calculation using the Kanjilal and Ghosh^[11] elasticity range suggests that an 81.7% pass-through rate would reduce demand by approximately 4.8 to 7.0% relative to the counterfactual of no hike, compared with the 5.9 to 8.6% reduction that full pass-through would have achieved. The difference is not trivial at the volumes India imports. Second, from the perspective of current account management, policymakers should expect that a portion of the intended import compression will be offset by growth in informal channels. Historical evidence suggests this offset is quantitatively significant: the pre-2024 15% duty regime coexisted with estimated unofficial imports well above 100 tonnes per year. A purely duty-based approach to gold import compression therefore operates against a structural smuggling elasticity that limits its effectiveness.

9 Conclusion

This paper estimates the pass-through of India’s May 2026 gold import duty hike to domestic wholesale gold prices using an interrupted time series design applied to a primary sample of 375

daily observations (346 pre-hike, 29 post-hike) drawn from 929 daily price observations (824 IBJA PDF circulars and 105 BullionWorld.in gap-fills) within a 1,170-row weekday date spine. The duty hike raised the domestic gold premium by Rs. 9,841 per 10 grams (Newey-West HAC SE Rs. 516, $N = 375$, Pre = 346, Post = 29), representing 81.7% of the theoretical ceiling of Rs. 12,039. Full pass-through is formally rejected ($t = -4.26$, $p < 0.001$). Local projections confirm the adjustment was immediate on the first trading day and held on a stable plateau through Day 14 (78.2%), with no anticipatory pricing. An ARDL bounds test (ARDL(4,3), $F = 9.005$, $N = 859$) confirms the pre-hike cointegrating relationship between domestic and international prices; the stable local-projection plateau through Day 14 is consistent with the shift persisting within the observation window. Placebo tests and counterfactual methods confirm the result is specific to May 13, 2026. The incomplete pass-through is consistent with supply-side dampening from front-loaded inventory imported at the prior duty rate, seasonal demand weakness, and historical smuggling elasticity.

These findings carry several limitations. First, the post-hike observation window is 29 trading days. The plateau observed through Day 14 may give way to further adjustment – toward or away from the full ceiling – as pre-hike inventory is depleted and informal supply channels adjust. Second, the IBJA benchmark is a wholesale price; the pass-through to retail jewellery consumers will differ depending on the competitive structure of the retail jewellery market and the extent to which distribution margins absorb or amplify the duty shock. Third, the paper does not directly observe import volumes or smuggling flows, so the decomposition of the absorption gap remains qualitative. A longer post-hike window will be decisive in distinguishing between a slow-adjustment narrative (the market is still finding its new equilibrium) and a structural-absorption narrative (supply-side mechanisms permanently limit pass-through under a 15% duty regime).

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